

CHALOCHALO

OCI Always Free + PAYG

Terraform + Application Architecture & Project Documentation

Area	Implemented direction
Cloud account	PAYG account; ChaloChalo OCI resources restricted to Always Free configurations.
Compute	VM.Standard.A1.Flex — 2 OCPU / 12 GB RAM.
Database	OCI MySQL.Free — 50 GB — private subnet.
Application	Responsive web UI + Nginx + FastAPI microservices + MySQL + Ollama.
Cost rule	Never use paid fallback for A1 capacity.

Updated edition includes the missing application architecture, service responsibilities, data flow and OCI deployment architecture.

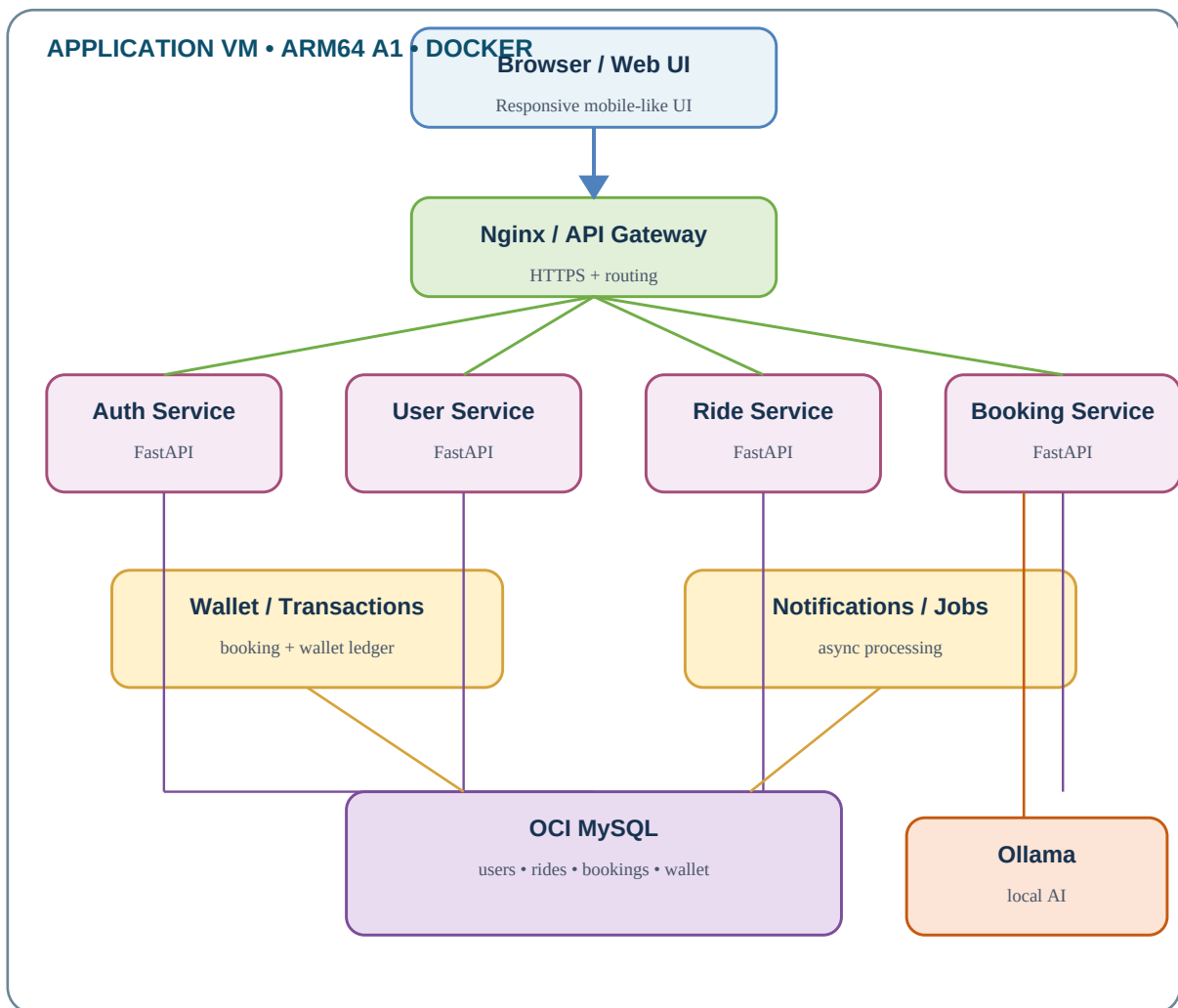
1. Executive Summary

ChaloChalo is an India-focused intercity carpool and daily office-commute platform. The application is designed as a responsive web application backed by FastAPI services, OCI MySQL and local Ollama AI. Terraform provisions the OCI infrastructure.

Cost/capacity policy: PAYG removes Free Tier account restrictions but does not guarantee A1 physical host capacity. A1 capacity errors are handled by changing availability domain in the home region or waiting—not by selecting paid compute.

2. Application Architecture

The application architecture below shows the logical request path independently from the OCI infrastructure. Docker hosts the application components on the ARM64 A1 VM.



3. Application Service Responsibilities

Service	Responsibility
Auth Service	Registration, login, token/session validation and password flows.

User Service	Profiles, user preferences and account information.
Ride Service	Publish ride, search ride, ride details, seat availability and ride lifecycle.
Booking Service	Seat reservation, cancellation and booking status lifecycle.
Wallet / Transaction	Wallet balance, booking-related transactions and ledger records.
Notification / Jobs	Async notifications, reminders and background processing.
Ollama	Local/private AI inference used by selected application features.

4. End-to-End Application Flow

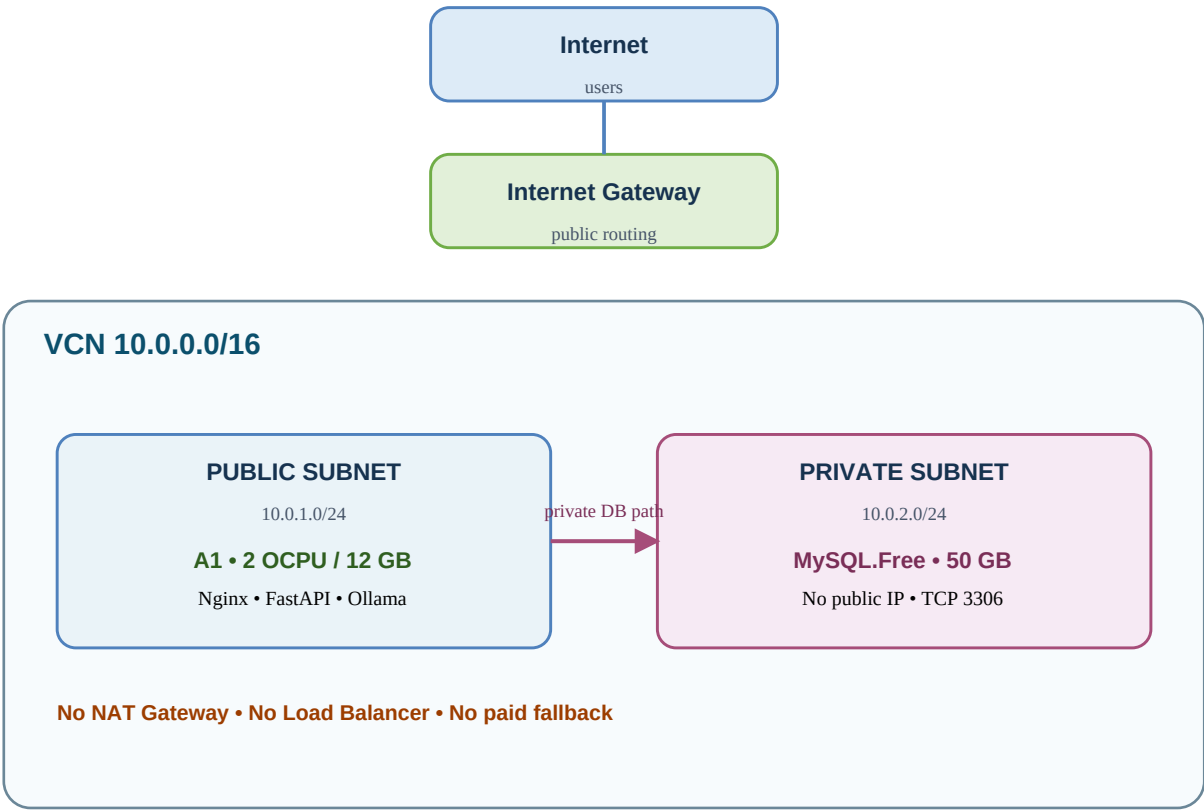
1. Browser sends HTTPS request to Nginx.
2. Nginx routes to the appropriate FastAPI service.
3. Auth validates identity/token where required.
4. Domain service applies business rules for ride, user, booking or wallet operation.
5. Service reads/writes OCI MySQL through the private network.
6. Optional AI operation calls Ollama locally on the VM.
7. JSON response returns through Nginx to the browser.

5. Data Architecture

Domain	Representative data
Identity	users, roles, profiles, sessions/tokens
Ride	rides, routes/stops, seats/availability
Booking	bookings, passengers, status history
Wallet	wallets, wallet transactions, payment references
Platform	notifications, audit logs, configuration

The table names above are the logical application model. Exact schema/columns should be finalized during MySQL implementation and migrations.

6. OCI Infrastructure Architecture



7. Terraform Structure & Core Policy

```
chalocho/
├── infrastructure/terraform/
│   ├── versions.tf
│   ├── variables.tf
│   ├── main.tf
│   ├── outputs.tf
│   ├── terraform.tfvars
│   └── .gitignore
```

7.1 Always Free constants

```
locals {
  compute_shape = "VM.Standard.A1.Flex"
  compute_ocpus = 2
  compute_memory_gb = 12
  mysql_shape = "MySQL.Free"
  mysql_storage_gb = 50
}
```

7.2 A1 instance selection

```
resource "oci_core_instance" "chalochalo" {
  compartment_id = var.compartment_ocid
  display_name = "chalochalo-server"
  availability_domain = var.a1_availability_domain
  shape = "VM.Standard.A1.Flex"

  shape_config {
    ocpus = 2
    memory_in_gbs = 12
  }
}
```

7.3 MySQL selection

```
resource "oci_mysql_mysql_db_system" "chalochalo" {
  compartment_id = var.compartment_ocid
  shape_name = "MySQL.Free"
  subnet_id = oci_core_subnet.private.id
  data_storage_size_in_gb = 50
  is_highly_available = false
}
```

8. Networking, Security & Secrets

Component	Configuration
VCN	10.0.0.0/16
Public subnet	10.0.1.0/24; public application VM
Private subnet	10.0.2.0/24; MySQL; public IP prohibited
Ingress	22 / 80 / 443 during bootstrap
Database ingress	TCP 3306 only from 10.0.1.0/24
Secrets	OCI key and MySQL password kept outside source control
SSH	Initial broad rule must be restricted after bootstrap

The OCI private key initially used in the setup was encrypted. A new unencrypted RSA key was generated so the OCI Terraform provider could load it directly. Keep it at chmod 600.

8.1 Redacted variables

```
private_key_path = "/Users/<user>/.oci/oci_api_key.pem"
region = "ap-mumbai-1"
mysql_admin_username = "chalo_admin"
a1_availability_domain = "ujaX:AP-MUMBAI-1-AD-1"
```

8.2 Secret environment variable

```
export TF_VAR_mysql_admin_password='YOUR_STRONG_PASSWORD'
echo ${TF_VAR_mysql_admin_password:+MYSQL_PASSWORD_SET}
```

9. PAYG + Always Free Guardrails

Rule	Policy
PAYG	Allowed as account model; does not mean paid resources are allowed.
A1	VM.Standard.A1.Flex, 2 OCPU / 12 GB.
MySQL	MySQL.Free, 50 GB.
Capacity	Try another home-region AD or wait; never increase to paid shape.
Extra services	No paid NAT/LB/Compute/Database services.
Terraform	Review plan before every apply.
Monitoring	Add OCI cost/usage monitoring as a second safety layer.

Target OCI infrastructure cost is \$0 while the implemented resources and usage remain within applicable Always Free allowances. PAYG can bill resources outside those allowances.

10. Deployment Workflow

10.1 Standard lifecycle

```
terraform init
terraform fmt
terraform validate
terraform plan -out=tfplan
terraform show -no-color tfplan
terraform apply "tfplan"
```

```
# Intentional decommission only:
terraform destroy
```

If the availability domain is changed after a plan is generated, delete/recreate the saved plan before applying.

11. Capacity Handling

The A1 Out of Capacity issue is a host-capacity issue, not something PAYG guarantees away. Terraform should make the A1 AD selectable.

11.1 AD variable

```
variable "a1_availability_domain" {
  type = string
}

# terraform.tfvars
a1_availability_domain = "ujaX:AP-MUMBAI-1-AD-1"

# instance
availability_domain = var.a1_availability_domain
```

Recommended future automation: pre-flight capacity check → AD-1 → AD-2 → AD-3 in the home region → select available AD → create only the 2 OCPU/12 GB A1 VM. If none is available, stop rather than falling back to paid capacity.

12. Ollama & AI Runtime

Ollama is planned as a local software service on the same A1 ARM64 VM. This avoids adding an external AI API service to the OCI architecture. The trade-off is limited CPU/RAM: model size, concurrency and context length must be kept appropriate for a 12 GB host.

AI flow	Implementation
Application → Ollama	Internal/local HTTP request from FastAPI service.
Model runtime	Ollama on A1 ARM64.
Database context	FastAPI may retrieve application data from private MySQL before AI processing.
Cost control	No separate OCI AI service added to the baseline.

13. Verification Checklist

Check	Expected
Provider	OCI authentication succeeds.
Compute	VM.Standard.A1.Flex / 2 OCPU / 12 GB.
Database	MySQL.Free / 50 GB / private.
Network	Public app subnet + private DB subnet.
MySQL	No public IP; 3306 limited to app subnet.
Paid resources	None in reviewed Terraform plan.
NAT/LB	Absent.
Application	Nginx → FastAPI services → MySQL; optional Ollama.
Secrets	Not committed to Git.

14. Known Limitations & Next Steps

- PAYG does not guarantee A1 host capacity.
- Add automated AD capacity pre-flight.
- Restrict SSH after bootstrap.
- Harden Terraform state and secret management.
- Finalize production MySQL schema and migrations.
- Add authentication/authorization, rate limiting, audit logging and observability.
- Tune Ollama model/concurrency for 2 OCPU / 12 GB.
- Add OCI cost/usage monitoring before production.

15. Final Architecture Summary

Layer	Final baseline
Product	Intercity carpool + daily office commute
Frontend	Responsive/mobile-like web application
Gateway	Nginx / API entry point
Backend	FastAPI microservices: Auth, User, Ride, Booking, Wallet/Jobs
Database	OCI MySQL.Free, 50 GB, private subnet
AI	Ollama on A1
Compute	VM.Standard.A1.Flex, 2 OCPU / 12 GB
Network	VCN 10.0.0.0/16 + public/private subnets
Cloud account	PAYG with Always Free resource policy
Capacity strategy	Home region; change AD or wait; no paid fallback

End of updated document • Generated 12 August 2026